

GRADE 7 COMPUTER SCIENCE

Working with MS Access 2016

A simple visual guide to databases, structures, and managing your tables.

BASED ON CHAPTER 1 TEXTBOOK GUIDELINES

SECTION 01

Foundations of Data

Understanding what databases are, why we use DBMS, and where they organize life.

| What is a **Database**?

The Simple Analogy

Imagine your school library. If books are scattered in messy piles, finding a book takes hours! But on organized shelves by subject, you can find it in seconds.

A **Database** is that digital shelf. A **DBMS** is the helpful librarian that keeps track of every single entry!



Database Definition

A structured collection of similar records or interrelated data stored systematically.



DBMS Definition

Software designed to store, modify, retrieve, and safeguard database values efficiently.

Popular **DBMS** Software



Oracle

Highly robust and powerful, engineered for enterprise-scale industrial networks and global websites.



MS-Access

Microsoft's Relational DBMS designed for personal use, desktop applications, and classroom projects.



SQL Server

A professional server database managed by Microsoft, engineered for high-performance scale.



Foxpro / Clipper

Classic flat and file-based systems that paved the early path for modern database technology.

Core DBMS Functions



Concurrency

Enables multiple users to access and work on the exact same database records concurrently without conflicts.



Security Control

Protects highly confidential files using secure password restrictions, preventing unauthorized modifications.



Backup & Recovery

Performs safe duplicate backups, ensuring your valuable records can be recovered if a hardware failure occurs.



Integrity Rules

Enforces strict structural rules to make sure entered values are correct, standardized, and perfectly clean.

Relational Databases

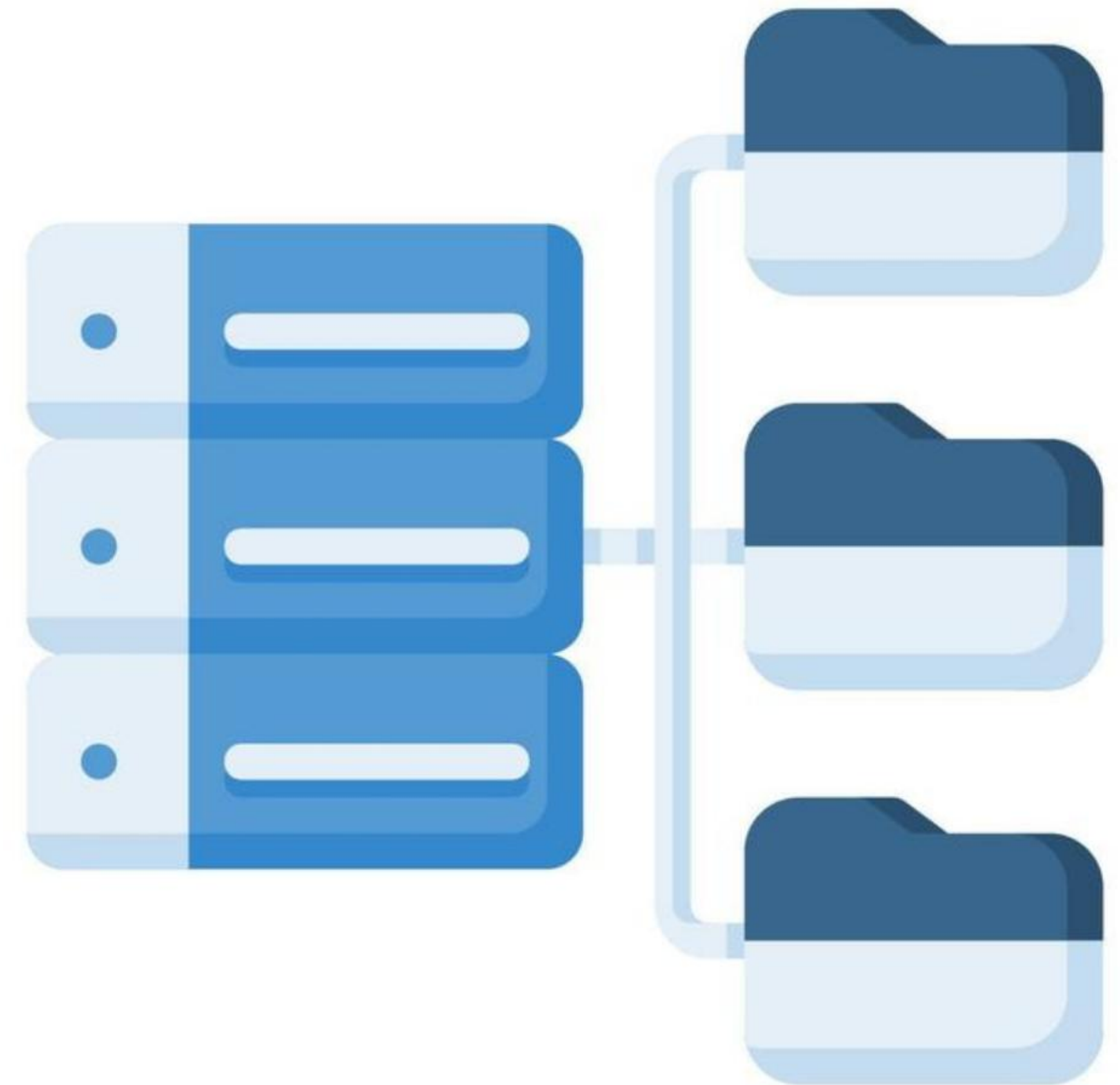
What is an RDBMS?

As discussed in the document [WhatsApp Image 2026-06-18 at 4.08.54 PM.jpeg](#), an RDBMS manages relationships between distinct tables, linking records using matched identifiers.

This critical mechanism eliminates redundant data storage (data redundancy) and guarantees seamless consistency across your records.

DATABASE ICON

Include icon based on 64px grid, High quality design, Editable size and pixel perfect



| Table Views in Access

Design vs. Datasheet

To construct your spreadsheet correctly, MS Access 2016 separates database architecture into two distinct operating modes:

-  **Design View:** Used to define fields, structural data properties, and format guidelines. Think of this as the blueprints!
-  **Datasheet View:** Used for real-time information entry in columns and rows. Think of this as the actual filled sheet!



Essential Field Data Types

Data Type	Description & Usage	Maximum Storage Limit
Short Text	Alphanumeric values (Names, addresses, flat text headers).	Up to 255 Characters
Long Text	Extended explanations, rich commentary text fields.	Up to 1 Gigabyte (GB)
Number	Pure numeric mathematical calculations (Ages, quantities).	1, 2, 4, or 8 Bytes
Date / Time	Specific operational calendar dates and hourly timestamps.	8 Bytes
Yes / No	Boolean conditions containing logical binary options (True/False).	1 Bit (0 or 1)
AutoNumber	Sequential increments serving as unique Primary Keys.	4 Bytes

Access Database Objects

As detailed in [WhatsApp Image 2026-06-18 at 4.08.55 PM \(2\).jpeg](#), an Access system is built using 4 core objects:



Tables

The core structural objects that store raw information in logical vertical columns and horizontal rows.



Queries

Used to quickly isolate and retrieve specific information matching pre-set condition rules from your tables.



Forms

User-friendly, custom layouts designed for inputting and viewing raw data records comfortably.



Reports

Formatted sheets prepared specifically for analytical printing and sharing key table takeaways.

| How to Create a Table



1. Open Access

Launch Access and click **Blank Desktop Database**.



2. Design View

Choose Design View and type in your Field headings.



3. Assign Types

Match fields to logical data formats (e.g. text, date).



4. Save & Fill

Name and save your table. Switch to Datasheet to type values!

Lab Activity: Employee Registry

Your assignment is to build this precise employee registry layout from **WhatsApp Image 2026-06-18 at 4.08.56 PM (1).jpeg**:

Emp ID (AutoNo)	Emp Name (Short Text)	Gender (Short Text)	Department (Short Text)	Designation (Short Text)
1	Rahul	M	Admin	Accounts Incharge
2	Sachin	M	Development	Software Engineer
3	Robin	M	Testing	Test Engineer
4	Rita Roy	F	IT Support	Sr. Executive
5	Jessy	F	IT Support	Project Manager

**Note: Enter total of 8 textbook employees. Refer to textbook page for full lists!*

| Lab Steps: **Action Plan**

04

Simple Lab Stages

- ✓ **Stage 1 – Setup:** Open MS Access. Select 'Blank Desktop Database'. Name your file as EmployeeDB.accdb.
- ✓ **Stage 2 – Define Fields:** Enter Design View. Add Emp ID, Emp Name, Gender, Department, and Designation as Short Text fields.
- ✓ **Stage 3 – Data Input:** Save the table structural layout. Switch views to Datasheet Mode. Type the employee rows accurately.
- ✓ **Stage 4 – Check Integrity:** Ensure that Emp ID acts as your unique Primary Key. Save and submit your database database project!

TIME TO EXPLORE

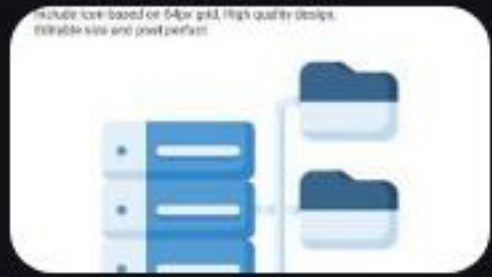
Any Questions?

Turn on your computer systems and launch Microsoft Access.
Let's design our very first database structure!



Practice Goal: Add 8 custom records from page 19 table.

Image Sources



https://static.vecteezy.com/system/resources/previews/003/565/319/non_2x/database-icon-with-flat-color-style-isolated-on-white-background-illustration-folder-server-sign-symbol-icon-concept-for-digital-it-logo-industry-technology-apps-web-and-project-vector.jpg

Source: www.vecteezy.com



[Thumbnail for https://static.vecteezy.com/system/resources/previews/050/812/725/non_2x/microsoft-office-programs-logo-icons-set-application-logos-editorial-word-excel-outlook-powerpoint-onenote-visual-studio-access-logo-collection-flat-style-microsoft-windows-products-free-vector.jpg](https://static.vecteezy.com/system/resources/previews/050/812/725/non_2x/microsoft-office-programs-logo-icons-set-application-logos-editorial-word-excel-outlook-powerpoint-onenote-visual-studio-access-logo-collection-flat-style-microsoft-windows-products-free-vector.jpg)

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